EXERCISES

Level 1

(Problems Based on Fundamentals)

1. Verify Rolle's theorem for the function
$f(x) = x^3 - 3x^2 + 2x + 5$ on $[0, 2]$..
2. Verify Rolle's theorem for the function
$f(x) = (x - a)^m (x - b)^n$ on $[a, b]$, $m, n \in I^+$
3. Verify Rolle's theorem for the function
$$f(x) = \log\left\{\frac{x^2 + ab}{x(a + b)}\right\} \text{ on } [a, b], \text{ where } 0 < a < b.$$
4. Verify Rolle's theorem for the function
$f(x) = \sin^4 x + \cos^4 x$ on $\left[0, \frac{\pi}{2}\right]$
5. Verify Rolle's theorem for the function
$f(x) = 2\sin x + \sin 2x$ on $[0, \pi]$
6. Verify Rolle's theorem for the function
$f(x) = \sin x + \cos x - 1$ on $\left[0, \frac{\pi}{2}\right]$
7. Verify Rolle's theorem for the function
$f(x) = e^x(\sin x - \cos x)$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$
8. Suppose that $f(x) = x^{1/2} - x^{3/2}$ on $[0, 1]$. Find a number c that satisfies the conditions of Rolle's theorem.
9. If the value of c prescribed in the Rolle's theorem for the function $f(x) = 2x(x - 3)^n$, $n \in N$ on $[0, 3]$ is $3/4$, then find the value of n.
10. If the function $f(x) = x^3 - 6x^2 + ax + b$ is defined on $[1, 3]$ satisfies the hypothesis of Rolle's theorem, then find the values of a and b.
11. If the function $f(x) = x^3 - px^2 + qx$ is defined on $[1, 3]$ satisfies the hypothesis of Rolle's theorem such that $c = 5/4$, then find the values of $(p + 4q + 17)$.
12. At what points on the curve
$y = 12(x + 1)(x - 2)$ on $[-1, 2]$, is the tangent parallel to x–axis?.

Algebraic Meaning of Rolle's Theorem

13. Let $f(x) = x^3 - x^2 - x + 1$. Prove that there is a root of its derivative on $(-1, 1)$.
14. Prove that the equation $x\cos x = \sin x$ has a root between π and 2π.
15. Prove that the equation $x^3 + x - 1 = 0$ has exactly one real root.
16. Let $f(x) = \sin^5 x + \cos^5 x - 1$ on, $\left[0, \frac{\pi}{2}\right]$
Then prove that the equation $f(x) = 0$ has two roots in $\left[0, \frac{\pi}{2}\right]$
17. If $2a + 3b + 6c = 0$, then prove that the equation $ax^2 + bx + c = 0$ has at least one root in $(0, 1)$.
18. If $a + b + c = 0$, then show that the equation $3ax^2 + 2bx + c = 0$ has atleast one root in $(0, 1)$.
19. If $3a + 4b + 6c + 12d = 0$, then prove that the equation $ax^3 + bx^2 + cx + d$, where $a, b, c, d \in R$ has atleast one root in $(0, 1)$.
20. If $f(x) = x^2(1 - x)^3$, then prove that the equation $f'(x) = 0$ has atleast one root in $(0, 1)$.
21. If $f(x) = (x - 1)(x - 2)(x - 3)(x - 4)$, then find the number of real roots of $f'(x) = 0$
and indicate the intervals in which they would lie.
22. Find the number of real roots of
(i) $x^3 - 6x^2 + 15x + 3 = 0$
(ii) $4x^2 - 21x^2 + 18x + 20 = 0$
(iii) $3x^4 - 8x^3 - 6x^2 + 24x + 1 = 0$
(iv) $x^4 - 4x - 2 = 0$

23. Discuss the applicability of Rolle's theorem for each of the following functions on the indicated intervals:

(i) $f(x) = |x - 2|$ in $[0, 3]$.

(ii) $f(x) = 3 + (x - 2)^{2/3}$ in $[1, 3]$

(iii) $f(x) = \sin\left(\frac{1}{x}\right)$ in $[-1, 1]$

(iv) $f(x) = \begin{cases} -4x + 5, 0 \le x \le 1 \\ 2x - 3, 1 < x \le 2 \end{cases}$.

(v) $f(x) = [x]$ in $[-1, 1]$, where [,] = G.I.F.

(vi) $f(x) = \sin|x|$ in $[-2, 2]$

(vii) $f(x) = e^{-|x|}$ in $[-3, 3]$

(viii) $f(x) = \left|e^{-|x|} - \frac{1}{2}\right|$ in $[-2, 2]$

(ix) $f(x) = \left|\frac{1}{x} - 1\right|$ in $[-1, 1]$

(x) $f(x) = \sin x + |\sin x|$ in $[-\pi, \pi]$

Lagranges Mean Value Theorem

24. Verify Lagranges Mean Value Theorem for the function $f(x) = x^3 - x^2 - x + 1$ on $[0, 2]$.

25. Find the value of c for which the function $f(x) = (x - 1)(x - 2)(x - 3)$ on $[0, 4]$ is applicable on L.M.V. theorem

26. Find the point on the curve $y = 2x^2 - 5x + 3$ where the tangent is parallel to the chord joining the points $A(1, 0)$ and $B(2, 1)$.

27. Find a point on the parabola $y = (x - 4)^2$, where the tangent is parallel to the chord joining $(4, 0)$ and $(5, 1)$.

28. Suppose $f(x)$ is twice differentiable function such that $f(1) = 1, f(2) = 4, f(3) = 9$, then prove that there exist at least one root in $(1, 3)$ such that $f''(x) = 2$, $x \in R$

29. If $f(x)$ satisfies the property of L.M.V. theorem in $[0, 2]$ such that $f(0) = 0$ and $f'(x) \le \frac{1}{2}$ for all x in $[0, 2]$, prove that $f(x) \le 1$.

30. Let $f(x)$ and $g(x)$ be differentiable functions for $x \in [0, 1]$ such that $f(0) = 2$, $g(0) = 0$, $f(1) = 6$ Let there exist a real number c in $(0, 1)$ such that $f'(c) = 2g'(c)$, then find the value of $g(1)$.

31. If $a, b, c \in \left(0, \frac{\pi}{2}\right)$ and $a < c < b$, then prove that $\frac{\cos a - \cos b}{\sin a - \sin b} = -\tan c$

32. Discuss the applicability of L.M.V. theorem for each of the following functions to the indicated intervals

(i) $f(x) = \frac{1}{x}$ on $[-1, 1]$

(ii) $f(x) = \sin|x|$ on $[-2\pi, 2\pi]$

(iii) $f(x) = |\sin x|$ on $[0, 2\pi]$

(iv) $f(x) = \log|x|$ on $[0, 5]$

(v) $f(x) = |\log|x||$ on $[-1, 1]$

33. Using L.M.V. theorem, prove that

$$\frac{b - a}{b} < \log\left(\frac{b}{a}\right) < \frac{b - a}{a},$$

where $0 < a < b$.

34. Using L.M.V. theorem, prove that $(b - a)\sec^2 a < \tan b - \tan a < (b - a)\sec^2 b$, where $0 < a < b < \frac{\pi}{2}$

35. Using L.M.V. theorem, prove that

$$\frac{\beta - \alpha}{1 + \beta^2} < \tan^{-1}\beta - \tan^{-1}\alpha < \frac{\beta - \alpha}{1 + \alpha^2}, \quad 0 < \alpha < \beta$$

36. Using L.M.V. theorem, prove that $|\sin x - \sin y| \le |x - y|$

37. Using L.M.V. theorem, prove that $|\tan^{-1} x| \le |x|$ for all x in R.

38. Using L.M.V. theorem, prove that $|\tan x - \tan y| \ge |x - y|$ for all x, y in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

39. Let f be a differentiable function for all x in R. If $f(1) = -2$ and $f'(x) \ge 2$ for all x in $[1, 6]$ such that the least value of $f(6)$ is m, then find the value of $2(m + 2)^3 + 17$

40. If f is continuous on $[0, 2]$ and differentiable on $(0, 2)$ such that $f(0) = 2$, $f(2) = 8$ and $f'(x) \le 3$ for all x in $(0, 2)$, then find the value of $f(1)$.

HINTS AND SOLUTIONS

Level I

(Problems based on Fundaments)

1. Given $f(x) = x^3 - 3x^2 + 2x + 5$

   As we know that, every polynomial function is continuous as well as differentiable.

   So, $f(x)$ is continuous and differentiable on the indicated interval.

   Also, $f(0) = 5$ and $f(2) = 8 - 12 + 4 + 5 = 5$

   i.e. $f(0) = 5 = f(2)$

   Thus, all the conditions of Rolle's theorem are satisfied.

   Now, we have to show that there exist a point c in $(0, 2)$ such that $f'(c) = 0$

   We have $f'(c) = 3c^2 - 6c + 2 = 0 = 0$ gives

   We have $f'(c) = 3c^2 - 6c + 2 = 0$ gives

   $$\Rightarrow \quad c = \frac{6 \pm \sqrt{36 - 24}}{6} = \frac{6 \pm 2\sqrt{3}}{6} = 1 \pm \frac{1}{\sqrt{3}}$$

   $$\Rightarrow \quad c = 1 \pm \frac{1}{\sqrt{3}} \in (0, 2)$$

   Hence, Rolle's theorem is verified.

2. Given $f(x) = (x - a)^m (x - b)^n$

   As we know that every polynomial function is continuous and differentiable everywhere.

   So, $f(x)$ is continuous and differentiable on the given indicated interval.

   Also, $f(a) = 0 = f(b)$.

   Thus, all the conditions of Rolle's theorem are satisfied.

   Now, we have to show that there exist a point c in (a, b) such that $f'(c) = 0$

   So,

   $f'(x) = m(x - a)^{m-1}(x - b)^n + n(x - a)^m(x - b)^{n-1}$

   $f'(x) = (x - a)^{m-1}(x - b)^{n-1}\,(m(x - b) + n(x - a))$

   Now, $f'(c)$ gives $c = a$, $c = b$ and

   $$\Rightarrow \quad (m(c - b) + n(c - a)) = 0$$

   $$\Rightarrow \quad c = \frac{mb + na}{m + n} \in (a, b)$$

   Hence, Rolle's theorem is verified.

3. Given $f(x) = \log\left\{\frac{x^2 + ab}{x(a + b)}\right\}$

   As we know that every logarithmic function is continuous and differentiable in Positive real numbers

   So it is continuous in $[a, b]$ and differentiable in (a, b)

   Also, $f(a) = \log\left(\frac{a^2 + ab}{a(a + b)}\right) = \log 1 = 0$

   and $f(b) = \log\left(\frac{b^2 + ab}{b(a + b)}\right) = \log 1 = 0$

   Thus, all the conditions of Rolle's theorem are satisfied.

   Now we have to show that there exist a point c in (a, b) such that $f'(c) = 0$

   We have $f(x) = \log\left(\frac{x^2 + ab}{x(a + b)}\right)$

   $$= \log(x^2 + ab) - \log(x(a + b))$$

   $$\Rightarrow \quad f'(x) = \frac{2x}{x^2 + ab} - \frac{1}{x}$$

   $$\Rightarrow \quad f'(x) = \frac{2x^2 - x^2 - ab}{x(x^2 + ab)} = \frac{x^2 - ab}{x(x^2 + ab)}$$

   Now, $f'(c) = 0$ gives $c^2 - ab = 0$

   $$\Rightarrow \quad c = \sqrt{ab} \in (a, b)$$

   Hence, Rolle's theorem is verified.

4. Given $f(x) = \sin^4 x + \cos^4 x$

   As we know that every sine and co-sine functions is continuous and differentiable everywhere.

So, it is continuous on $\left[0, \frac{\pi}{2}\right]$ and differentiable on $\left(0, \frac{\pi}{2}\right)$

Also, $\quad f(0) = 1 = f\left(\frac{\pi}{2}\right)$

Thus, all the conditions of Rolle's theorem are satisfied.

Now, we have to show that there exist a point c in $\left(0, \frac{\pi}{2}\right)$ such that $f'(c) = 0$

Therefore,

$$\begin{aligned} f'(x) &= 4\sin^3 x\cos x - 4\cos^3 x\sin x \\ &= 4\sin x\cos x(\sin^2 x - \cos^2 x) \\ &= -2.(2\sin x\cos x)(\cos^2 x - \sin^2 x) \\ &= -2.\sin 2x\cdot\cos 2x \\ &= -\sin 4x \end{aligned}$$

Now, $\quad f'(c) = 0$ gives $\sin(4c) = 0$

$\Rightarrow \quad \sin(4c) = \sin(\pi)$

$\Rightarrow \quad c = \frac{\pi}{4} \in \left(0, \frac{\pi}{2}\right).$

5. Clearly, $f(x)$ is continuous in $[0, \pi]$ and differentiable in $(0, \pi)$

$f(0) = 0, f(\pi) = 0$

$\Rightarrow \quad f(0) = 0 = f(\pi)$

Thus, all the conditions of Rolle's theorem are satisfied.

Now, we have to show that, there exists a point $c \in (0, \pi)$ such that $f'(c) = 0$.

Now, $\quad f'(c) = 0$ gives

$\Rightarrow \quad 2\cos c + 2\cos 2c = 0$

$\Rightarrow \quad \cos c + \cos 2c = 0$

$\Rightarrow \quad \cos c + 2\cos^2 c - 1 = 0$

$\Rightarrow \quad 2\cos^2 c + \cos c - 1 = 0$

$\Rightarrow \quad 2\cos^2 c + 2\cos c - 1 = 0$

$\Rightarrow \quad 2\cos c(\cos c + 1) - (\cos c + 1) = 0$

$\Rightarrow \quad (2\cos c - 1)(\cos c + 1) = 0$

$\Rightarrow \quad \cos c = \frac{1}{2}, -1$

$\Rightarrow \quad c = \frac{\pi}{3}, \pi$

$\Rightarrow \quad c = \frac{\pi}{3} \in [0, \pi]$

Thus, the Rolle's theorem is verified.

6. As we know that, every sine and cosine function is differentiable. So, $f(x)$ is continuous in $\left[0, \frac{\pi}{2}\right]$ and differentiable in $\left(0, \frac{\pi}{2}\right)$

$f(0) = 0 + 1 - 1 = 0$

$f\left(\frac{\pi}{2}\right) = 1 + 0 - 1 = 0$

So, $\quad f(0) = 0 = f\left(\frac{\pi}{2}\right)$

Thus, all the conditions of Rolle's theorem are satisfied.

Now, we have to show that, there exists a point $c \in \left(0, \frac{\pi}{2}\right)$ such that $f'(c) = 0$.

$\Rightarrow \quad \cos c - \sin c = 0$

$\Rightarrow \quad \cos c = \sin c$

$\Rightarrow \quad \tan c = 1$

$\Rightarrow \quad c = \frac{\pi}{4} \in \left(0, \frac{\pi}{2}\right)$

Thus, the Rolle's theorem is verified.

7. As we know that, every exponential, sine and cosine functions are differentiable everywhere.

So, it is continuous in $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$

and differentiable in $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$

Now, $\quad f\left(\frac{\pi}{4}\right) = e^{\frac{\pi}{4}}\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) = 0$

and $\quad f\left(\frac{5\pi}{4}\right) = e^{\frac{5\pi}{4}}\left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right) = 0$

So, $\quad f\left(\frac{\pi}{4}\right) = 0 = f\left(\frac{5\pi}{4}\right)$

Thus, all the conditions of Rolle's theorem are satisfied.

Now, we have to show that, there exists a point $c \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ such that $f'(c) = 0$.

$\Rightarrow \quad e^c(\sin c - \cos c) + e^c(\cos c + \sin c) = 0$

$\Rightarrow \quad (\sin c - \cos c) + (\cos c + \sin c) = 0$

$\Rightarrow \quad 2\sin c = 0$

$\Rightarrow \quad \sin c = 0$

$\Rightarrow \quad c = \pi \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$

Thus, all the conditions of Rolle's theorem are verified.

8. Given $\quad f(x) = x^{1/2} - x^{3/2}$ on $[0, 1]$

Since $f(x)$ satisfies all the conditions of Rolle's theorem, so there exists a point $c \in (0, 1)$ such that

$$f'(c) = 0$$

$$\Rightarrow \quad \frac{1}{2}c^{-1/2} - \frac{3}{2}c^{1/2} = 0$$

$$\Rightarrow \quad \frac{1}{c^{1/2}} - 3c^{1/2} = 0$$

$$\Rightarrow \quad 1 - 3c = 0$$

$$\Rightarrow \quad c = \frac{1}{3} \in (0, 1)$$

9. Given $\quad f(x) = 2x(x-3)^n$

$\Rightarrow \quad f'(x) = 2(x-3)^n + 2nx(x-3)^{n-1}$

Now, $\quad f'(c) = 0$ gives

$\Rightarrow \quad 2(c-3)^n + 2nc(c-3)^{n-1} = 0$

$\Rightarrow \quad 2(c-3)^n = -2nc(c-3)^{n-1}$

$\Rightarrow \quad (c-3)^n = -nc(c-3)^{n-1}$

$\Rightarrow \quad (c-3) = -nc$

$\Rightarrow \quad c(1+n) = 3$

$\Rightarrow \quad (1+n) = \frac{3}{c} = \frac{3}{3/4} = 4$

$\Rightarrow \quad n = 3$

Hence, the value of n is 3.

10. Given $\quad f(x) = x^3 - 6x^2 + ax + b$

$f(1) = f(3)$

$\Rightarrow \quad 1 - 6 + a + b = 27 - 54 + 3a + b$

$\Rightarrow \quad a + b - 5 = 3a + b - 27$

$\Rightarrow \quad 2a = 22$

$\Rightarrow \quad a = 11,\ b \in R$

11. Given $\quad f(x) = x^3 - px^2 + qx$

$f(1) = f(3)$

$\Rightarrow \quad 1 - p + q = 27 - 9p + q$

$\Rightarrow \quad 1 - p = 27 - 9p$

$\Rightarrow \quad 8p = 26$

$\Rightarrow \quad p = \frac{13}{4}$

Also, $\quad f'(c) = 0$

$\Rightarrow \quad 3c^2 - 2pc + q = 0.$

$\Rightarrow \quad 3\left(\frac{5}{4}\right)^2 - 2p\left(\frac{5}{4}\right) + q = 0$

$\Rightarrow \quad q = 2p\left(\frac{5}{4}\right) - 3\left(\frac{5}{4}\right)^2$

$\Rightarrow \quad q = 2\left(\frac{13}{4} \times \frac{5}{4}\right) - \frac{75}{16} = \frac{130 - 75}{16} = \frac{55}{16}$

Hence, the value of $(p + 4q + 17)$

$$= \frac{13}{4} + \frac{55}{4} + 17$$

$$= \frac{68}{4} + 17 = 34$$

12. Given curve is $y = 12(x+1)(x-2)$

Since the tangent is parallel to x–axis, so

so, $\quad \frac{dy}{dx} = 0$

$\Rightarrow \quad 12\{(x+1) + (x-2)\} = 0$

$\Rightarrow \quad 2x - 1 = 0$

$\Rightarrow \quad x = \frac{1}{2}$

when $x = 1/2$, then $y = 12\left(1 + \frac{1}{2}\right)\left(\frac{1}{2} - 2\right)$

$\Rightarrow \quad y = 12 \times \frac{3}{2} \times -\frac{3}{2} = -27$

Hence, the point is $\left(\frac{1}{2}, -27\right)$.

13. Given $\quad f(x) = x^3 - x^2 - x + 1$

$\Rightarrow \quad f'(x) = 3x^2 - 2x - 1$

$f(x) = 0$ gives $3x^2 - 2x - 1 = 0$

$\Rightarrow \quad 3x^2 - 3x + x - 1 = 0$

$\Rightarrow \quad 3x(x-1) + 1(x-1) = 0$

$\Rightarrow \quad (3x+1)(x-1) = 0$

$\Rightarrow \quad x = 1, -\frac{1}{3}$

$\Rightarrow \quad x = -\frac{1}{3} \in (-1, 1)$

Thus, the derivative of the function has a root in $(-1, 1)$.

14. Here, we shall show that the equation $x\cos x = \sin x$ has a root in π and 2π.

Let $\quad f(x) = \frac{\sin x}{x}$

Clearly $\quad f(x) = 0$, at $x = \pi, 2\pi$

Now, $\quad f'(x) = \frac{x\cos x - \sin x}{x^2}$

So, by Rolle's theorem, $f'(x)$ must vanish for some value of x in$(\pi, 2\pi)$.

Hence, the equation $x\cos x - \sin x = 0$ has a root in $(\pi, 2\pi)$.

15. Let $\quad f(x) = x^3 + x - 1$

Now, $\quad f(0) = 0 + 0 - 1 = -1 < 0$

$f(1) = 1 + 1 - 1 = 1 > 0$

So, $f(x)$ has a root in between 0 and 1.

16. Given $\quad f(x) = \sin^5 x + \cos^5 x - 1$

$f'(x) = 5\sin^4 x\cos x - 5\cos^4 x\sin x$

$= 5\sin x\cos x(\sin^3 x - \cos^3 x)$

Now, $\quad f'(x) = 0$ gives

$\Rightarrow \quad \sin x\cos x(\sin^3 x - \cos^3 x) = 0$

$\Rightarrow \quad \tan^3 x = 1$

$\Rightarrow \quad x = \frac{\pi}{4}$

Since $f'(x)$ has a real root, so $f(x)$ must have two roots in $\left[0, \frac{\pi}{2}\right]$

17. Let $\quad f(x) = \frac{ax^3}{3} + \frac{bx^2}{2} + cx$

Now $\quad f(0) = 0$

and $\quad f(1) = \frac{a}{3} + \frac{b}{2} + c = \frac{2a + 3b + 6c}{6} = \frac{0}{6} = 0$

Thus, $ax^2 + bx + c = 0$ has at-least one root in (0, 1).

18. Let $\quad f(x) = ax^3 + bx^2 + cx$

Now, $\quad f(0) = 0$

and $\quad f(1) = a + b + c = 0.$

Thus, by Rolle's Theorem, there is atleast one root in its derivative.

19. Let $\quad f(x) = \frac{ax^4}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + dx$

Now, $\quad f(0) = 0$

and $\quad f(1) = \frac{a}{4} + \frac{b}{3} + \frac{c}{2} + d$

$$= \frac{3a + 4b + 6c + 12d}{12} = 0$$

So, by Rolle's theorem, between any two roots of a polynomial, there is atleast one root of its derivative.

Hence, the result.

20. Given $\quad f(x) = x^2(1 - x)^3$

Clearly, $\quad f(0) = 0 = f(1)$

Thus, $\quad f(x)$ has two roots.

So, by Rolle's theorem, between any two roots of a polynomial, there is atleast one root of its derivative.

Hence, the result.

21. Given $\quad f(x) = (x - 1)(x - 2)(x - 3)(x - 4)$

$f'(x) = (x - 1)(x - 2)(x - 3) + (x - 1)(x - 3)(x - 4) + (x - 1)(x - 2)(x - 4) + (x - 2)(x - 3)(x - 4)$

$= (x^3 - 6x^2 + 11x - 6) + (x^3 - 8x^2 + 19x - 12) + (x^3 - 7x^2 + 14x - 8) + (x^3 - 9x^2 + 26x - 24)$

$= 4x^3 - 30x^2 + 70x - 50$

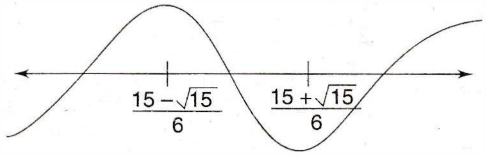
$f''(x) = 12x^2 - 60x + 70$

Clearly, its $D = 3600 - 3360 = 240 > 0$

So, $f''(x)$ has two real roots.

Hence, $f'(x) = 0$ has three real roots.

Now, $f''(x) = 0$ gives $x = \frac{15 \pm \sqrt{15}}{6}$



Thus, $f'(x) = 0$ has three real roots

$\left(-\infty, \frac{15 - \sqrt{15}}{6}\right), \left(\frac{15 - \sqrt{15}}{6}, \frac{15 + \sqrt{15}}{6}\right)$

and $\quad \left(\frac{15 + \sqrt{15}}{6}, \infty\right)$

22.

(i) Let $f(x) = x^3 - 6x^2 + 15x + 3$

$f'(x) = 3x^2 - 12x + 15$

$= 3(x^2 - 4x + 5)$

Now, $f'(x) = 0$ has no real roots

So, $f(x) = 0$ has only one real root.

(ii) Let $f(x) = 4x^3 - 21x^2 + 18x + 20$

$\Rightarrow f'(x) = 12x^2 - 42x + 18$

$\Rightarrow f'(x) = 6(2x^2 - 7x + 3)$

$\Rightarrow f'(x) = 6(2x - 1)(x - 3)$

Clearly, $f'(x) = 0$ has two real roots.

Thus, $f(x) = 0$ has three real roots.

(iii) Let $f(x) = 3x^4 - 8x^3 - 6x^2 + 24x + 1$

$f'(x) = 12x^3 - 24x^2 - 12x + 24$

$= 12(x^3 - 2x^2 - x + 2)$

$= 12\{x^2(x - 2) - 1(x - 2)\}$

$= 12(x - 2)(x^2 - 2)$

Clearly, $f'(x) = 0$ has three real roots.

So, $f(x) = 0$ has 4 real roots.

(iv) Let $f(x) = x^4 - 4x - 2$

$f'(x) = 4x^3 - 4$

$= 4(x^3 - 1)$

$= 4(x - 1)(x^2 + x + 1)$

So, it has only one real root

Thus, $f(x) = 0$ has 2 real roots.

23.

(i) $f(x) = |x - 2|$ in [0, 3].

Clearly, $f(x)$ is not differentiable at $x = 2$

So Rolle's theorem is not applicable.

(ii) $f(x) = 3 + (x - 2)^{2/3}$ in $[1, 3]$

Clearly, $f(x)$ is not differentiable at $x = 2$ So Rolle's theorem is not applicable.

(iii) $f(x) = \sin\left(\frac{1}{x}\right)$ in $[-1, 1]$

Clearly, $f(x)$ is not continuous at $x = 0$ So Rolle's theorem is not applicable.

(iv) $f(x) = \begin{cases} -4x + 5, 0 \le x \le 1 \\ 2x - 3, 1 < x \le 2 \end{cases}$.

Clearly, $f(x)$ is not continuous at $x = 1$ So Rolle's theorem is not applicable.

(v) $f(x) = [x]$ in $[-1, 1]$, where [,] = G.I.F.

Clearly, $f(x)$ is not continuous at $x = -1, 0, 1$ So Rolle's theorem is not applicable.

(vi) $f(x) = \sin|x|$ in $[-2, 2]$

Clearly, $f(x)$ is not differentiable at $x = 0$

So Rolle's theorem is not applicable.

(vii) $f(x) = e^{-|x|}$ in $[-3, 3]$

Clearly, $f(x)$ is not differentiable at $x = 0$

So Rolle's theorem is not applicable.

(viii) $f(x) = \left|e^{-|x|} - \frac{1}{2}\right|$ in $[-2, 2]$

Clearly, $f(x)$ is not differentiable at $x = 0$

So Rolle's theorem is not applicable.

(ix) $f(x) = \left|\frac{1}{x} - 1\right|$ in $[-1, 1]$

Clearly, $f(x)$ is not continuous at $x = 0$

So Rolle's theorem is not applicable.

(x) $f(x) = \sin x + |\sin x|$ in $[-\pi, \pi]$

Clearly, $f(x)$ is not differentiable at $x = 0$

So Rolle's theorem is not applicable.

Lagranges Mean Value Theorem

24. Given $f(x) = x^3 - x^2 - x + 1$

As we know that, every polynomial function is continuous and differentiable everywhere.

So, it is continuous in [0, 2] and differentiable in (0, 2)

Thus, all the conditions of L.M.V. theorem are satisfied

Now, we have to show that, there exists a point c in (0, 2) such that $f'(c) = \frac{f(2) - f(0)}{2 - 0)}$

$$\Rightarrow \quad f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{3 - 1}{2} = 1$$

$$\Rightarrow \quad 3c^2 - 2c - 1 = 1$$

$$\Rightarrow \quad 3c^2 - 2c - 2 = 0$$

$$\Rightarrow \quad c = \frac{2 \pm \sqrt{4 + 24}}{6} = \frac{2 \pm 2\sqrt{7}}{6}$$

$$\Rightarrow \quad c = \frac{1 \pm \sqrt{7}}{3}$$

$$\Rightarrow \quad c = \frac{1 + \sqrt{7}}{3} \in (0, 2)$$

Hence, L.M.V. theorem is verified.

25. Given $f(x) = (x - 1)(x - 2)(x - 3)(x - 4)$

As we know that every polynomial function is continuous and differentiable everywhere.

So, it is continuous in [0, 4] and differentiable in (0, 4)

Thus, all the conditions of L.M.V. theorem are satisfied.

Now we have to show that there exists a point $c \in (0, 4)$ such that

$$f'(c) = \frac{f(4) - f(0)}{4 - 0} = \frac{24 - 0}{4} = 6$$

$$\Rightarrow \quad 4c^3 - 30c^2 + 70c - 50 = -6$$

$$\Rightarrow \quad 4c^3 - 30c^2 + 70c - 44 = 0$$

$$\Rightarrow \quad 2(c - 1)(2c^2 - 13c + 11) = 0$$

$$\Rightarrow \quad 2(c - 1)^2(2c - 11) = 0$$

$$\Rightarrow \quad c = 1, \frac{11}{2}$$

$$\Rightarrow \quad c = 1 \in (0, 4)$$

Hence, Lagranges Mean Value Theorem is verified.

26. Given curve is $y = 2x^2 - 5x + 3$

The arc AB is continuous in [1, 2] and differentiable in (1, 2)

So, by L.M.V. Theorem,

$$f'(x) = \frac{f(2)f(1)}{2 - 1} = f(2) - f(1)$$

$$\Rightarrow \quad 4x - 5 = 1 - 0$$

$$\Rightarrow \quad 4x = 6$$

$$\Rightarrow \quad x = \frac{3}{2}$$

when $x = \frac{3}{2}$, then $y = 0$

Thus, the point is $\left(\frac{3}{2}, 0\right)$

27. Since the tangent is parallel to the chord joining (4, 0) and (5, 1), so we get,

$$2(x - 4) = \frac{1 - 0}{5 - 4} = 1$$

$$(x - 4) = \frac{1}{2}$$

$$x = 4 + \frac{1}{2} = \frac{9}{2}$$

when $x = \frac{9}{2}$, then $y = \left(\frac{9}{2} - 4\right)^2 = \frac{1}{4}$

Hence, the point is $\left(\frac{9}{2}, \frac{1}{4}\right)$

28. Let $g(x) = f(x) - x^2$

$\Rightarrow$ $g(x)$ has atleast 3 real roots which are $x = 1, 2$ and 3

$\Rightarrow$ $g'(x)$ has at least 2 real roots in $x \in (1, 3)$

$\Rightarrow$ $g''(x)$ has at least 1 real root in $x \in (1, 3)$

$\Rightarrow$ $f''(x) = 2$ for at least one root in $x \in (1, 3)$

Hence, the result.

29. Applying L.M.V. theorem, we have,

$$f'(x) = \frac{f(x) - f(0)}{2 - 0} = \frac{f(x)}{2}$$

$$\Rightarrow \quad \frac{f(x)}{2} = f'(x) \le \frac{1}{2}$$

$$\Rightarrow \quad f(x) \le 1$$

Hence, the result.

30. For the function $f(x)$, applying L.M.V. Theorem,

we get, $f'(c) = \dfrac{f(1) - f(0)}{1 - 0} = f(1) - f(0)$ ...(i)

Also, for the function $g(x)$, we have,

$$g'(c) = \frac{g(1) - g(0)}{1 - 0} = g(1) - g(0) \quad \text{...(ii)}$$

Dividing (i) and (ii), we get,

$$\frac{f'(c)}{g'(c)} = \frac{f(1) - f(0)}{g(1) - g(0)}$$

$$\Rightarrow \quad 2 = \frac{6 - 2}{g(1) - 0} = \frac{4}{g(1)}$$

$$\Rightarrow \quad g(1) = 2$$

31. For the function $f(x) = \cos x$,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\Rightarrow \quad -\sin c = \frac{\cos b - \cos a}{b - a} \quad \text{...(i)}$$

For the function $f(x) = \sin x$,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\Rightarrow \quad f'(c) = \frac{\sin b - \sin a}{b - a} \quad \text{...(ii)}$$

Dividing (i) and (ii), we get,

$$\frac{\cos b - \cos a}{\sin b - \sin a} = \frac{\sin c}{\cos c} = -\tan c$$

32.

(i) $f(x) = \frac{1}{x}$ on $[-1, 1]$

Clearly, $f(x)$ is not continuous at $x = 0$

So, L.M.V. theorems is not applicable.

(ii) $f(x) = \sin |x|$ on $[-2\pi, 2\pi]$

Clearly, $f(x)$ is not differentiable at $x = 0$

So, L.M.V theorems is not applicable.

(iii) $f(x) = |\sin x|$ on $[0, 2\pi]$

Clearly, $f(x)$ is not continuous at $x = \pi$

So, L.M.V. theorems is not applicable.

(iv) $f(x) = \log|x|$ on $[0, 5]$

Clearly, $f(x)$ is not continuous at $x = 0$

So, L.M.V. theorems is not applicable.

(v) $f(x) = |\log|x||$ on $[-1, 1]$

Clearly, $f(x)$ is not continuous at $x = -1, 0, 1$

So, L.M.V. theorems is not applicable.

33. Let $f(x) = \log x$, $0 < a < b$

Clearly, $f(x)$ is continuous and differentiable in (a, b), where $0 < a < b$

$$f'(x) = \frac{1}{x}$$

$$f'(c) = \frac{1}{c}, \quad a < c < b$$

By L.M.V, theorem,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$= \frac{\log\left(\frac{b}{a}\right)}{b - a}$$

Now, $a < c < b$

$$\Rightarrow \quad \frac{1}{b} < \frac{1}{c} < \frac{1}{a}$$

$$\Rightarrow \quad \frac{1}{b} < \frac{\log\left(\frac{b}{a}\right)}{b - a} < \frac{1}{a}$$

$$\Rightarrow \quad \frac{b - a}{b} < \log\left(\frac{b}{a}\right) < \frac{b - a}{a}$$

Hence, the result.

34. Let $f(x) = \tan x$ in (a, b), where $0 < a < b < \frac{\pi}{2}$

Clearly, $f(x)$ is continuous in $[a, b]$ and in (a, b)

Now, $f'(x) = \sec^2 x$

$\Rightarrow \quad f'(c) = \sec^2 c$

By L.M.V Theorem,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$f'(c) = \frac{\tan b - \tan a}{b - a}$$

Here, $\quad c \in (a, b)$

$\Rightarrow \quad a < c < b$

$\Rightarrow \quad \sec^2 a < \sec^2 c < \sec^2 b$

$$\Rightarrow \quad \sec^2 a < \frac{f(b) - f(a)}{b - a} < \sec^2 b$$

$$\Rightarrow \quad \sec^2 a < \frac{\tan b - \tan a}{b - a} < \sec^2 b$$

$\Rightarrow \quad (b - a)\sec^2 a < (\tan b - \tan a) < (b - a)\sec^2 b$

Hence, the result.

35. Let $\quad f(x) = \tan^{-1}x$ in (a, b), $0 < \alpha < \beta$

Clearly $f(x)$ is continuous and differentiable in (α, β)

$$f'(x) = \frac{1}{1 + x^2}$$

$$\Rightarrow \quad f'(c) = \frac{1}{1 + c^2}$$

Now, $\quad \alpha < c < \beta$

$\Rightarrow \quad \alpha^2 < c^2 < \beta^2$

$\Rightarrow \quad 1 + a^2 < 1 + c^2 < 1 + \beta^2$

$$\Rightarrow \quad \frac{1}{1 + \beta^2} < \frac{1}{1 + c^2} < \frac{1}{1 + \alpha^2}$$

$$\Rightarrow \quad \frac{1}{1 + \beta^2} < f'(c) < \frac{1}{1 + \alpha^2}$$

$$\Rightarrow \quad \frac{1}{1 + \beta^2} < \frac{f(\beta) - f(\alpha)}{\beta - \alpha} < \frac{1}{1 + \alpha^2}$$

$$\Rightarrow \quad \frac{\beta - \alpha}{1 + \beta^2} < f(\beta) - f(\alpha) < \frac{\beta - \alpha}{1 + \alpha^2}$$

$$\Rightarrow \quad \frac{\beta - \alpha}{1 + \beta^2} < \tan^{-1}\beta - \tan^{-1}\alpha < \frac{\beta - \alpha}{1 + \alpha^2}$$

Hence, the result.

36. Let $f(x) = \sin x$

Clearly, $f(x)$ is continuous and differentiable So L.M.V. theorem is applicable.

$$f'(x) = \cos x$$

$\Rightarrow \quad f'(c) = \cos c$

$$\Rightarrow \quad \frac{f(y) - f(x)}{y - x} = \cos c$$

$$\Rightarrow \quad \frac{f(x) - f(y)}{x - y} = \cos c$$

$$\Rightarrow \quad \left|\frac{f(x) - f(y)}{x - y}\right| = |\cos c|$$

$$\Rightarrow \quad \left|\frac{\sin x - \sin y}{x - y}\right| \le 1$$

$\Rightarrow \quad |\sin x - \sin y| \le |x - y|$

Hence, the result.

37. Let $f(x) = \tan^{-1}x - x$ in $[0, x]$

Clearly, $f(x)$ is continuous and differentiable. So L.M.V. theorem is applicable.

$$f'(x) = \frac{1}{1 + x^2} - 1 = -\frac{x^2}{1 + x^2}$$

$$\Rightarrow \quad f'(c) = -\frac{c^2}{1 + c^2}$$

$$\Rightarrow \quad \frac{f(x) - f(0)}{x - 0} = -\frac{c^2}{1 + c^2}$$

$$\Rightarrow \quad \frac{f(x)}{x} = -\frac{c^2}{1 + c^2}$$

$$\Rightarrow \quad \frac{f(x)}{x} = -\frac{c^2}{1 + c^2} < 0$$

$$\Rightarrow \quad \frac{\tan^{-1}x - x}{x} \le 0$$

$\Rightarrow \quad \tan^{-1}x - x \le 0$

$\Rightarrow \quad \tan^{-1}x \le x$

$\Rightarrow \quad |\tan^{-1}x| \le |x|$

Hence, the result.

38. Let $f(x) = \tan x$

Clearly, $f(x)$ is continuous and differentiable for all x, y in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

So, L.M.V. theorems is applicable.

$\Rightarrow \quad f'(x) = \sec^2 x$

$\Rightarrow \quad f'(c) = \sec^2 c$

$$\Rightarrow \quad \frac{f(x) - f(y)}{x - y} = \sec^2 c$$

$$\Rightarrow \quad \frac{\tan x - \tan y}{x - y} = \sec^2 c$$

$\Rightarrow \quad \left|\frac{\tan x - \tan y}{x - y}\right| = |\sec^2 c|$

$\Rightarrow \quad \left|\frac{\tan x - \tan y}{x - y}\right| \geq 1$

$\Rightarrow \quad |\tan x - \tan y| \geq |x - y|$

Hence, the result.

39. Given that f is differentiable in (1, 6) So by L.M.V Theorems

$$\frac{f(6) - f(1)}{6 - 1} = f'(x)$$

$\Rightarrow \quad \frac{f(6) - f(1)}{5} \geq 2$

$\Rightarrow \quad f(6) - f(1) \geq 10$

$\Rightarrow \quad f(6) \geq f(1) + 10$

$\Rightarrow \quad f(6) \geq -2 + 10 = 8$

Thus, $\quad m = 8$

Now, the value of $2(m + 2)^3 + 12$

$$= 2(8 + 2)^3 + 17$$

$$= 2017.$$

40. When f is continuous in [0, 1] and differentiable in (0, 1)

So by L.M.V theorem, we get,

$$f'(x) = \frac{f(1) - f(0)}{1 - 0}$$

$\Rightarrow \quad f(1) - f(0) = f'(x)$

$\Rightarrow \quad f(1) - 2 \leq 3$

$\quad f(1) \leq 5$ ...(i)

When f is continuous in [1, 2] and differentiable in (1, 2).

So, by L.M.V. theorem, we get,

$$f'(x) = \frac{f(2) - f(1)}{2 - 1}$$

$\Rightarrow \quad f(2) - f(1) = f'(x)$

$\Rightarrow \quad f(2) - f(1) \leq 3$

$\Rightarrow \quad 8 - f(1) \leq 3$

$\Rightarrow \quad f(1) \geq 5$ ...(ii)

From (i) and (ii), we get,

$$f(1) = 5$$